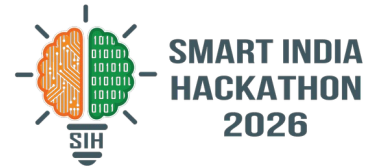
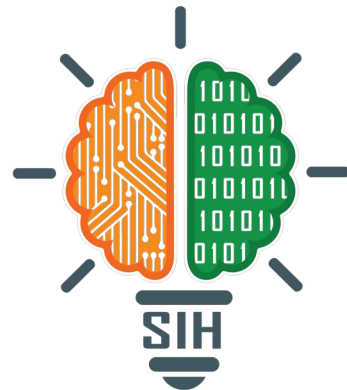


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – SIH26097
- Problem Statement Title – AI-driven voice assistant for livelihood mapping and NSQF-aligned skilling recommendations for SC communities under the GIA component of PM-AJAY
- Theme – Agriculture, FoodTech & Rural Development
- PS Category – Software
- Team ID – _____ (fill from the SIH portal)
- Team Name (Registered on portal) – Cognify



SMART INDIA HACKATHON 2026

Proposed Solution · one free phone call that turns a person's skills into the right work



Why this needs fixing SIH26097 · MoSJE

PM-AJAY pays for skill training and small-business support (up to ₹50,000) for poor SC families. But the course is chosen through paper forms and field officers — people who cannot read or do not know the trades land in the wrong course, drop out, and get no job.

41%

of trained people actually got a job

Source: CAG audit 2025

34%

of SC adults cannot read

Source: Census 2011

36%

of women have no phone of their own

Source: NFHS-6, 2023-24

ONE CALL, SIX STEPS — ABOUT 10 MINUTES, ON ANY PHONE



1

Missed call → call-back

Free call-back within a minute from an official 1600 govt number, in the language he picks.



2

Facts by keypad

Age, education, distance, job-or-business pressed on keys — a speech error can't change eligibility.



3

He talks, AI listens

"Aap kya kaam karte ho?" AI maps it to an official NCO-2015 occupation and asks about his skills.



4

Read back & saved

Everything is read back; press 1 to confirm. Saved as one clean table row, not a paper form.



5

1-4 options that fit him

Courses, schemes or jobs near him that truly fit — each fact checked against govt data. He can ask any doubt.



6

Choose → CSC → officer

Press 1-4. SMS with Case ID + nearest CSC; officer approves in the AJAY app; follow-up calls at 1, 3, 6 months.

WHAT ONLY SKILLCALL DOES — BEYOND THE SIX STEPS



Smart questions

Each question depends on the last answer: a bike mechanic is asked about engine repair, tyres, oil service.



Full profile, auto-built

The AI writes the complete profile and case table in the proper format by itself. Officers only review — hours of staff time saved per case.



Only government data

Answers come only from data we load: scheme rules, NQR courses, NCO jobs, centres. No internet, nothing invented.



Validator AI — SATYA

Before any fact is spoken, a second checker matches it to a database row. No match → the sentence is dropped.



Cheapest AI first

Small model by default; a bigger one only if unsure or the caller is unhappy; a human calls back if still stuck.



Web portal too

Mobile + OTP login. The person, CSC or officer can view and edit the same profile anytime — or skip the call entirely.

Also answers the 5 GIA issues in the brief → planning · beneficiary selection · placement after training · coordination · too few ground staff

TECH STACK

Calls & channels

WhatsApp Business API Exotel / Plivo • calls + SMS

Voice engine & AI models

Pipecat voice server Saaras v3 • speech → text
Bulbul v3 • text → speech Sarvam-30B / 105B • LLM
IndicConformer (free STT) Bhashini • MeitY
SATYA validator (ours)

Backend & data

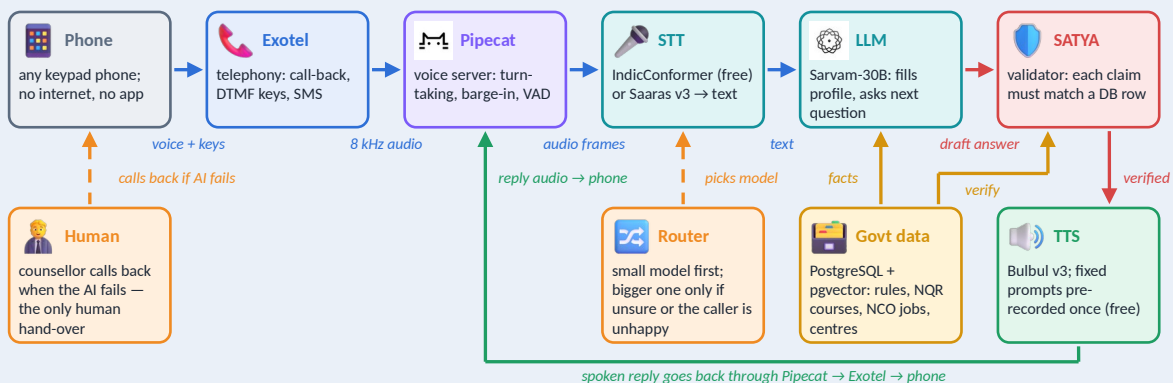
Python FastAPI
Celery Redis
PostgreSQL pgvector • semantic search

Apps, hosting & govt links

React portal + dashboard MeghRaj govt cloud*
NCO-2015 • occupation codes NQR / NCVE • courses
AJAY app* • officer workflow

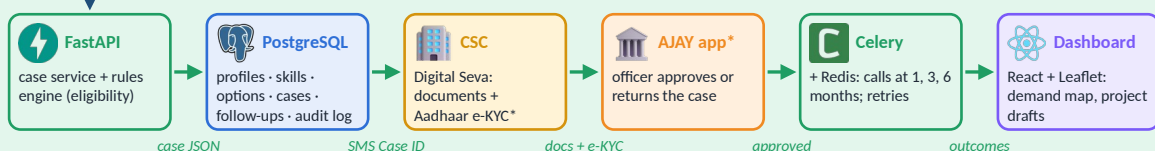
HOW IT WORKS — ARCHITECTURE

① LIVE CALL — ~1.5 s per turn



call ends → confirmed profile (JSON) → FastAPI

② AFTER THE CALL — the confirmed profile becomes a case



Web portal (React): login with mobile number + OTP → the same FastAPI and the same tables. The family, the CSC operator or the officer can view and edit the case at any time — no phone call needed.

AI understands → verified data decides → SATYA checks → humans approve

Data rules: no Aadhaar number stored; raw audio deleted after confirmation; consent logged (DPDP Rules 2025); every edit in the audit log.

* needs a government MoU.

Full architecture, API docs and code → [GitHub repo \(add link\)](#)

Can it be built and run? Yes — every piece already works in India today, every model is available by API now, and one completed call costs about ₹32–65.

ALREADY WORKING IN INDIA — NOTHING HERE IS NEW TECHNOLOGY



10 of 22 languages

UIDAI already runs Sarvam's voice AI for Aadhaar in 10 of its 22 Indian languages, on its own servers.

Source: PIB / MeitY 2025



12 lakh calls a day

Kilkari, the government's health-advice call system, already handles this volume every day.

Source: BMJ Global Health 2021



≤ 20% word error

Best Indian speech AI measured on real phone calls: 36,691 speakers, 15 languages.

Source: Voice of India benchmark 2026



All parts available today

Sarvam APIs, Pipecat, Exotel trial, PostgreSQL, React — the prototype can be built right now.

Source: our prototype plan

WHAT ONE COMPLETED CALL COSTS — ₹ PER PERSON

Completed call = 10-min interview + options + 4 follow-up calls (≈ 18 min of phone time).

Cost item	How we calculate it	Low	High
Phone line (Exotel)	18 min × ₹0.80–1.80 per minute	₹16.5	₹37.6
Speech → text (Saaras v3)	₹30 per hour of audio	₹9.4	₹9.5
Text → speech (Bulbul v3)	₹3 per 1,000 characters, spoken sentences only	₹2.4	₹6.0
AI brain (Sarvam-30B / 105B)	~90% of turns on 30B; 105B only when needed	₹0.3	₹1.5
SMS + human counsellor	4–6 SMS; 10–20% of callers need a human call-back	₹2.6	₹7.5
Hosting & helpline number	servers, database, number — spread per call	₹1.0	₹3.0
TOTAL per completed profile	1,000 missed calls → 517 finished profiles	≈ ₹32	≈ ₹65

Falls further over time: common questions get pre-written, verified answers (cached), so the AI thinks less each call. Unit prices: Sarvam & Exotel pricing pages; funnel is our estimate.

RISKS AND HOW WE HANDLE THEM

Village dialects are hard → facts by keypad · read back · bigger model, then a human

AI could say something wrong → SATYA blocks anything not in the govt database

People ignore unknown numbers → he calls first; call-back from an official 1600 number

WHO GETS THROUGH — AND WHO NEEDS A HUMAN

- 50% answer on the 1st attempt, 76% by the 3rd, 99.5% by the 9th
- ~80% of answered calls finish the interview; ~85% of those complete the options
- 10–20% need a human counsellor call-back — design target ≤ 20%
- Cannot be reached by phone? The web portal (mobile + OTP) is always open

Source: Kilkari, BMJ Global Health 2021

Source: our funnel estimate

Source: our design target

CAN THE GOVERNMENT AFFORD IT?



0.04–0.07% of the yearly budget A whole state costs ₹0.8–1.6 crore a year; PM-AJAY's budget is ₹2,140 crore a year.



₹9–13 lakh to start 6-month pilot in one block (≈ 1,000 families): build, test with real users, then scale.



Nothing new to build on the ground 6.5 lakh CSC centres and the AJAY officer app already exist — SkillCall plugs into them.

IMPACT AND BENEFITS

Who gains? Poor SC families, CSC operators, district officers and the Ministry. Every number below shows how it was worked out.

WHAT CHANGES



More trained people get jobs

41% → 70%

Only 41% of trained people get a job today; 70% is the government's own target. Per 1,000 trainees, even 50% means 90 more people working.

Source: CAG 2025 · MSDE norms



More people stay in the job

+11 percentage points

e.g. from 41% to 52% still in the job — when a trainee is told the real pay and place before joining. SkillCall tells him on the call.

Source: Chakravorty et al. 2024 (field trial)



Officers get their time back

40 h → 7 h a month

Per officer handling 40 people a month: before $40 \times 60 \text{ min} = 40 \text{ h}$; after $40 \times 10 \text{ min} \approx 7 \text{ h} \rightarrow 33 \text{ h saved}$. India-wide: 50,000 GIA cases $\times 50 \text{ min} \approx 41,700 \text{ h a year} \approx 24$ full-time officers.

Source: our estimate



Less money wasted

₹12.5 crore a year

If just 5 in 100 wrong matches are avoided: 2,500 people $\times$ ₹50,000 support — 19–39 times what SkillCall costs to run.

Source: our estimate

WHO WINS



SC family

Right course the first time, free, in his own language — and a follow-up call until he is actually earning.



CSC operator

The case arrives pre-filled as a table; only documents and e-KYC are left to do.



District officer

Ready, checked cases in the AJAY app instead of a 40-minute survey; a demand map by block for the annual plan.



Ministry

One record from call to job; every outcome marked confirmed (employer / EPFO*) or self-reported.

BENEFITS



Social

Women and non-readers included; free, in his own language; right course the first time.



Economic

Less money spent on wrong training; more people earning — proved by follow-up, not assumed.



Governance

One shared case record for CSC, officer, partner and bank; planning data instead of paper forms.



Environment

No paper forms, fewer trips to offices, runs on the shared government cloud.



Government data

CAG Report No. 20 of 2025 — PMKVY audit (41% placed) cag.gov.in

PM-AJAY GIA norms & beneficiaries — PIB 2023 / 2025 pib.gov.in

PM-AJAY budget ₹2,140 cr — Demands for Grants 2026-27 indiabudget.gov.in

NFHS-6 (women's phone access) - Census 2011 (SC literacy) pib.gov.in



Technology

Sarvam AI — Saaras v3, Bulbul v3, Sarvam-30B / 105B, pricing sarvam.ai

AI4Bharat IndicConformer — open Indian speech recognition ai4bharat.iitm.ac.in

Pipecat — open-source voice-agent framework pipecat.ai

Bhashini (MeitY) · Exotel · PostgreSQL · FastAPI · React bhashini.gov.in



Research papers

Chakravorty et al. 2024, J. Dev. Econ. — honest job info → +11 pts retention [sciencedirect.com](https://www.sciencedirect.com)

Bashingwa et al. 2021, BMJ Global Health — Kilikari call answer rates gh.bmj.com

Voice of India 2026 — Indian speech-AI benchmark on real calls arxiv.org

Medhi et al. 2011, ACM TOCHI — voice UIs for low-literacy users dl.acm.org



Standards & policy

NCO-2015 occupation codes · NQR / NCVET job roles nqr.gov.in

DPDP Rules 2025 — consent and data protection meitv.gov.in

TRAI 1600-series rules for government calls (2026) traigov.in

UIDAI-Sarvam voice AI deployment — PIB / MeitY 2025 pib.gov.in

SEE MORE — EVERYTHING THE SIX SLIDES COULD NOT FIT



GitHub repo

full code, architecture docs, API reference

github.com/cognify/skillcall [\(add link\)](#)



Live prototype

try a SkillCall conversation from your browser

[\(add link\)](#)



Demo video

2-minute walkthrough of a real call

[\(add link\)](#)



Reference file

all 16 parts, every number with its working

[\(add link\)](#)